

COVARIANCE DECAY FOR GAUSSIAN QUEUES

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ABSTRACT. We study large-lag cross-covariances of stationary workloads in two Gaussian queues driven by a common input and drained at possibly different rates. For standard Brownian input, we derive an exact representation in terms of a drifted Brownian hitting time and obtain the sharp asymptotic $K_{12}r^{-3/2}e^{-c_2^2r/2}$, with an explicit positive constant for all service rates. For fractional Brownian input with $H \neq 1/2$, we obtain the signed polynomial asymptotic $H(2H-1)\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}r^{2H-2}$ where τ_{W_i} is the maximizing time of $B_H(t) - c_it$ over $[0, \infty)$. For integrated stationary Gaussian input whose covariance is regularly varying, we show that the workload covariance inherits the same scale. We also derive an explicit exponential asymptotic for integrated Ornstein–Uhlenbeck input under the condition $c_2 > 2\sigma$. The proofs combine Gaussian Price interpolation with localization and tail estimates for the maximizing times.

1. INTRODUCTION

Gaussian queues have attracted substantial interest over the past several decades, particularly in the analysis of stationary buffer content. Although the steady-state marginal distribution is now well understood, much less is known about the temporal dependence of the workload process.

Consider an infinite-buffer queue driven by $X(t)$, $t \in \mathbb{R}$, a Gaussian process with stationary increments and continuous sample paths and drained at a constant rate $c > 0$. Let $W(t) = X(t) - ct$, $t \in \mathbb{R}$ and assume that $W(t) \rightarrow -\infty$ almost surely as $t \rightarrow \infty$. With the interpretation that $X(t) - X(s)$ is the amount of *fluid* that entered the system in time interval $[s, t]$ for $t > s$ the stationary workload process $Q_W(t)$, $t \in \mathbb{R}$ admits the representation

$$Q_W(t) := \sup_{u \geq 0} (W(t) - W(t - u)), \quad t \in \mathbb{R};$$

see, e.g., [12]. Modelling the input stream in a fluid queue by a Gaussian process is supported by functional central limit theorems; see e.g., [14] or the monograph [15].

An important class of models is obtained by taking $W(t) = B_H(t) - ct$, where B_H is a two-sided fractional Brownian motion (fBm) with Hurst parameter $H \in (0, 1)$ and $c > 0$. This defines the classical stationary storage model fed by fractional Brownian motion. Statistical evidence of *long range dependence* and *self-similarity* in input streams motivates this class of models [13]; see also [10].

One formulation of the temporal-dependence question highlighted in [5] asks for the large-time behaviour of

$$\rho(t) = \text{Corr}(Q_W(t), Q_W(0)), \quad t \rightarrow \infty,$$

and more generally whether the dependence range of the Gaussian input is inherited by the workload process. The answer depends sharply on the input. The Brownian case $H = 1/2$ is Markovian and exceptional. The covariance of the stationary reflected Brownian queue decays exponentially. It was shown in [1] that

$$\text{Cov}(Q_W(0), Q_W(t)) \sim \frac{2^{3/2}}{\sqrt{\pi} c^5} t^{-3/2} e^{-c^2 t/2}, \quad t \rightarrow \infty.$$

The above asymptotic follows from the representation

$$(1) \quad \text{Cov}(Q_W(0), Q_W(t)) = \int_0^\infty s e^{-2cs} \mathbb{P}\{\tau_s > t\} ds, \quad t > 0,$$

where τ_s is the hitting time of zero by a Brownian motion with drift $-c$ started from s .

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For fractional Brownian input with $H \neq 1/2$, no Markovian representation such as (1) is available. As we show below the covariance decays polynomially rather than exponentially. The key observation is that the cross-covariance of two distant increments of lengths u and v is asymptotic to $H(2H-1)uvr^{2H-2}$. The *Price interpolation* transfers this leading kernel to the two maximizing locations and yields the required asymptotic of the buffer content covariance.

We also treat integrated stationary Gaussian input

$$W(t) = \int_0^t Z(s) ds - ct,$$

where Z is a centered stationary Gaussian process. Under the assumptions stated below if its covariance satisfies $r_Z(r) \sim \kappa r^{-\alpha} L(r)$ as $r \rightarrow \infty$, then the workload covariance has the same regularly varying scale $r^{-\alpha} L(r)$, again with a constant determined by independent maximizing locations of the limiting one-dimensional variational problem. We shall consider integrated Ornstein–Uhlenbeck input separately and show that its exponential covariance decay is inherited by the workload under a sufficient condition on the service rate.

The method also handles cross-covariances of two queues driven by the same input.

Organisation of the paper. The main results are stated in Section 2; the auxiliary interpolation, compactness, uniqueness, and tail estimates are collected in Section 3; the proofs are presented in Section 4.

2. MAIN RESULTS

Let

$$W_i(t) = X(t) - c_i t, \quad c_i > 0, \quad i = 1, 2,$$

where X is a centered Gaussian process with stationary increments, continuous sample paths and $X(0) = 0$ almost surely. For a continuous process U on $[0, \infty)$ we write

$$M_U(T) := \sup_{0 \leq s \leq T} U(s), \quad \tau_U(T) := \arg \max_{0 \leq s \leq T} U(s)$$

and write M_U, τ_U for the corresponding maximum and maximizer over $[0, \infty)$ whenever the maximizer is unique. We begin with Brownian input and use the Markov property of the queue. Let $B(t), t \in \mathbb{R}$ be a standard two-sided Brownian motion with $B(0) = 0$ and take $X = B$. For $a \geq 0$ and $c > 0$ we set

$$\tau_a^{(c)} := \inf\{u \geq 0 : a + B(u) - cu = 0\}$$

and

$$U_i := Q_{W_i}(0), \quad m_i := \mathbb{E}\{U_i\} = \frac{1}{2c_i}, \quad i = 1, 2.$$

Theorem 2.1. *For every $T \geq 0$*

$$(2) \quad \text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \int_0^\infty \mathbb{E}\{(U_1 - m_1)\mathbf{1}_{\{U_2 > a\}}\} \mathbb{P}\{\tau_a^{(c_2)} > T\} da.$$

Moreover as $r \rightarrow \infty$ we have

$$(3) \quad \text{Cov}(Q_{W_1}(0), Q_{W_2}(r)) \sim K_{12} r^{-3/2} e^{-c_2^2 r/2},$$

where

$$0 < K_{12} = \sqrt{\frac{2}{\pi}} \begin{cases} \frac{2(2c_2 - c_1)}{c_2^6}, & 0 < c_1 \leq c_2, \\ \frac{2}{c_2^5} \left(1 - \frac{c_1 - c_2}{\sqrt{(c_1 - c_2)^2 + c_2^2}} \right), & c_1 > c_2. \end{cases}$$

Remark 2.2. At a fixed time the vector $(Q_{W_1}(0), Q_{W_2}(0))$ has the same law as

$$\left(\sup_{u \geq 0} (B(u) - c_1 u), \sup_{u \geq 0} (B(u) - c_2 u) \right).$$

Thus it is a two-point instance of the stationary maximum process studied in [7]. In particular their covariance formula yields

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(0)) = \frac{1}{4c_+^2} \left(2 - \frac{c_-}{c_+} \right), \quad c_+ = \max(c_1, c_2), \quad c_- = \min(c_1, c_2)$$

while the main focus of this contribution concerns the temporal lag covariance $\text{Cov}(Q_{W_1}(0), Q_{W_2}(r))$ as $r \rightarrow \infty$.

We next consider $X = B_H$ with $H \in (0, 1) \setminus \{1/2\}$ and normalization

$$\mathbb{E}\{B_H(t)\} = 0, \quad \text{Cov}(B_H(s), B_H(t)) = \frac{1}{2}(|s|^{2H} + |t|^{2H} - |t-s|^{2H}).$$

By construction each Q_{W_i} is stationary with the same 1-dimensional law as M_{W_i} .

Theorem 2.3. *As $r \rightarrow \infty$ we have*

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(r)) \sim H(2H-1)\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}r^{2H-2}.$$

The preceding result also identifies the dependence range of the stationary workload. For every $H \neq 1/2$ we have

$$\text{Corr}(Q_{W_1}(0), Q_{W_2}(r)) \sim \frac{H(2H-1)\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}}{\sqrt{\text{Var}(M_{W_1})\text{Var}(M_{W_2})}}r^{-2(1-H)}, \quad r \rightarrow \infty.$$

For $H > 1/2$ the workload has positive non-integrable correlation. If $H < 1/2$ the correlation is eventually negative and integrable and the workload is short-range dependent in the usual absolute-integrability sense. We now consider integrated stationary Gaussian input. Let $Z(t)$, $t \in \mathbb{R}$ be a centered stationary Gaussian process with continuous sample paths and covariance

$$r_Z(t) := \text{Cov}(Z(0), Z(t)).$$

The input process is

$$X(t) := \int_0^t Z(s) ds,$$

with the usual sign convention for $t < 0$.

Theorem 2.4. *Let the function L be slowly varying and eventually positive. If for some $\kappa \neq 0$ and $\alpha \geq 0$*

$$(4) \quad r_Z(t) \sim \kappa t^{-\alpha} L(t), \quad t \rightarrow \infty$$

and the additional condition $\lim_{t \rightarrow \infty} L(t) \log t = 0$ holds when $\alpha = 0$, then we have

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(r)) \sim \kappa r^{-\alpha} L(r) (\mathbb{E}\{\tau_{W_1}\} \mathbb{E}\{\tau_{W_2}\}), \quad r \rightarrow \infty.$$

Finally we consider integrated Ornstein–Uhlenbeck input. Let $Z(t)$, $t \in \mathbb{R}$ be a centered stationary Ornstein–Uhlenbeck Gaussian process with covariance

$$r_Z(t) := \text{Cov}(Z(0), Z(t)) = \sigma^2 e^{-\lambda|t|}, \quad \sigma > 0, \quad \lambda > 0.$$

Theorem 2.5. *If Z is an Ornstein–Uhlenbeck process as above and $c_2 > 2\sigma$, then we have*

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(r)) \sim \frac{\sigma^2}{\lambda^2} e^{-\lambda r} \mathbb{E}\{1 - e^{-\lambda \tau_{W_1}}\} \mathbb{E}\{e^{\lambda \tau_{W_2}} - 1\}, \quad r \rightarrow \infty.$$

Remark 2.6. (1) The regular-variation assumption in (4) is understood subject to positive definiteness of r_Z . In particular not every formal choice of κ , α and L is admissible as a covariance tail.

(2) In the above results the moment factors appearing in the covariance asymptotics are positive; the sign of the full asymptotic constant is determined by the corresponding covariance-kernel coefficient such as $H(2H-1)$ or κ . Namely, we have

$$\mathbb{E}\{\tau_W\} > 0, \quad \text{Var}(M_W) > 0.$$

Moreover for every $T > 0$ we have $\mathbb{E}\{\tau_W(T)\} > 0$. Indeed the moment estimates in Lemmas 3.11 and 3.12 together with the one-dimensional estimates established at the beginning of the proof of Theorem 2.5 imply in the respective cases that

$$M_W \in L^2, \quad \tau_W \in L^1.$$

For every $u > 0$ we have

$$\mathbb{V}\text{ar}(X(u)) > 0, \quad W(u) = X(u) - cu.$$

For fractional Brownian input this is immediate. For the regularly varying integrated input it follows from the spectral non-degeneracy argument in Lemma 3.12. For integrated Ornstein–Uhlenbeck input the explicit variogram satisfies

$$V_X(u) = \frac{2\sigma^2}{\lambda}u - \frac{2\sigma^2}{\lambda^2}(1 - e^{-\lambda u}) > 0, \quad u > 0.$$

Thus $W(u)$ is a non-degenerate Gaussian random variable and

$$\mathbb{P}\{W(u) > 0\} > 0.$$

Consequently $\mathbb{P}\{M_W > 0\} > 0$. On the event $\{M_W > 0\}$ the unique maximizer cannot be 0 because $W(0) = 0$. Hence we obtain

$$\mathbb{P}\{\tau_W > 0\} > 0, \quad \mathbb{E}\{\tau_W\} > 0.$$

For every $x \in \mathbb{R}$ we also have

$$\mathbb{P}\{M_W > x\} \geq \mathbb{P}\{W(u) > x\} > 0,$$

so M_W is not almost surely constant. Since $M_W \in L^2$ it follows that $\mathbb{V}\text{ar}(M_W) > 0$. The same argument with any $u \in (0, T]$ yields

$$\mathbb{E}\{\tau_W(T)\} > 0, \quad T > 0.$$

In particular in the integrated Ornstein–Uhlenbeck case we have

$$\mathbb{E}\{1 - e^{-\lambda\tau_W}\} > 0, \quad \mathbb{E}\{e^{\lambda\tau_W} - 1\} > 0$$

whenever the second expectation is finite.

- (3) For a single drift c the sufficient condition $c > 2\sigma$ for the integrated Ornstein–Uhlenbeck process has a natural large-deviation interpretation. For integrated OU input we have

$$\mathbb{V}\text{ar} X(t) = \frac{2\sigma^2}{\lambda}t - \frac{2\sigma^2}{\lambda^2}(1 - e^{-\lambda t}) \sim \frac{2\sigma^2}{\lambda}t, \quad t \rightarrow \infty.$$

On long time scales $X(t)$ behaves like Brownian motion with variance parameter $2\sigma^2/\lambda$ and

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}\{X(t) > ct\} = -\iota(c),$$

where

$$\iota(c) = \frac{c^2\lambda}{4\sigma^2}.$$

The limiting Price kernel in the integrated OU case contains $e^{\lambda\tau_W} - 1$ so our transfer argument uses $\mathbb{E}\{e^{\lambda\tau_W}\} < \infty$. The Borell–TIS estimate implies

$$\mathbb{E}\{e^{q\tau_W}\} < \infty \quad \text{for every } q < \iota(c).$$

In particular $q = \lambda$ is admissible whenever $c > 2\sigma$.

3. AUXILIARY RESULTS

3.1. Finite-dimensional and compact Price interpolation. The finite-dimensional identity below is the two-block covariance form of Price’s theorem; see [11, 16]. We establish the compact and half-line extensions needed here, including the treatment of nonunique maximizing sets, because these forms are not immediate from the standard finite-dimensional statement.

Lemma 3.1. *Let $(\xi, \eta) = ((\xi_i)_{i \in I}, (\eta_j)_{j \in J})$ be a centered Gaussian vector indexed by finite non-empty sets I, J . Put*

$$R_{ij} := \text{Cov}(\xi_i, \eta_j).$$

Let ξ' and η' be independent centered Gaussian vectors with the same marginal laws as ξ and η , independent of (ξ, η) , and set

$$\xi^{(\theta)} := \sqrt{\theta} \xi + \sqrt{1-\theta} \xi', \quad \eta^{(\theta)} := \sqrt{\theta} \eta + \sqrt{1-\theta} \eta'.$$

If $F \in C^1(\mathbb{R}^I)$ and $G \in C^1(\mathbb{R}^J)$ have bounded first partial derivatives, then

$$\text{Cov}(F(\xi), G(\eta)) = \int_0^1 \sum_{i \in I} \sum_{j \in J} R_{ij} \mathbb{E}\{\partial_i F(\xi^{(\theta)}) \partial_j G(\eta^{(\theta)})\} d\theta.$$

Proof. First suppose that the joint covariance matrix of (ξ, η) is positive definite and that $F, G \in C^2$ have bounded first and second derivatives. The standard Gaussian interpolation form of Price's theorem, applied to

$$H(\theta) := \mathbb{E}\{F(\xi^{(\theta)})G(\eta^{(\theta)})\}$$

yields

$$H'(\theta) = \sum_{i \in I} \sum_{j \in J} R_{ij} \mathbb{E}\{\partial_i F(\xi^{(\theta)}) \partial_j G(\eta^{(\theta)})\}.$$

Indeed, the marginal covariance blocks are independent of θ , while the derivative of the cross-covariance block is (R_{ij}) . Integration over $[0, 1]$ yields the formula. At fixed positive-definite covariance, the same identity holds for C^1 functions with bounded first derivatives by mollification. The mollified gradients remain uniformly bounded and converge locally uniformly, while the mollified functions have uniform linear growth, so dominated convergence applies on both sides.

It remains to remove the non-degeneracy assumption. Let U, V, U', V' be mutually independent standard Gaussian vectors of the appropriate dimensions, independent of all previously defined vectors, and put

$$\xi_\varepsilon := \xi + \sqrt{\varepsilon} U, \quad \eta_\varepsilon := \eta + \sqrt{\varepsilon} V,$$

with independent marginal copies

$$\xi'_\varepsilon := \xi' + \sqrt{\varepsilon} U', \quad \eta'_\varepsilon := \eta' + \sqrt{\varepsilon} V'.$$

The covariance matrix of $(\xi_\varepsilon, \eta_\varepsilon)$ is the covariance matrix of (ξ, η) plus εId , and is therefore positive definite, whereas its cross-covariance block is still (R_{ij}) . Hence the asserted identity holds with $\xi_\varepsilon, \eta_\varepsilon$ in place of ξ, η . On this common probability space we have

$$\xi_\varepsilon^{(\theta)} \rightarrow \xi^{(\theta)}, \quad \eta_\varepsilon^{(\theta)} \rightarrow \eta^{(\theta)}$$

almost surely for every θ . Since bounded first derivatives make F and G Lipschitz, $F(\xi_\varepsilon^{(\theta)}) \rightarrow F(\xi^{(\theta)})$ and $G(\eta_\varepsilon^{(\theta)}) \rightarrow G(\eta^{(\theta)})$ in L^2 , so the covariance converges. Continuity and boundedness of the first derivatives ensure convergence of the integrand on the right, dominated uniformly on $[0, 1] \times \Omega$. Letting $\varepsilon \downarrow 0$ proves the result for arbitrary, possibly singular, Gaussian covariance matrices.

This completes the proof. $\square$

Hereafter E and F are two non-empty compact metric spaces. For a given $h \in C(E)$ and $D_m \subset E$ we introduce the *softmax* probability measure

$$\pi_{m,\lambda}^h := \sum_{u \in D_m} \frac{e^{\lambda h(u)}}{\sum_{z \in D_m} e^{\lambda h(z)}} \delta_u.$$

Lemma 3.2. *Given $h \in C(E)$ set $A := \arg \max_{u \in E} h(u)$ and let $D_m \subset E$ be nonempty finite sets such that*

$$\Delta_m := \sup_{x \in E} d(x, D_m) \rightarrow 0, \quad m \rightarrow \infty.$$

For every $\varphi \in C(E)$ that is constant on A , with $\varphi|_A = c$, we have

$$\lim_{m \rightarrow \infty} \lim_{\lambda \rightarrow \infty} \int_E \varphi(u) \pi_{m,\lambda}^h(du) = c.$$

Proof. For fixed m , let

$$A_m := \arg \max_{u \in D_m} h(u).$$

Since D_m is finite and nonempty, A_m is finite and nonempty. As $\lambda \rightarrow \infty$, the softmax weights concentrate uniformly on A_m . Hence

$$\lim_{\lambda \rightarrow \infty} \int_E \varphi(u) \pi_{m,\lambda}^h(du) = \frac{1}{|A_m|} \sum_{u \in A_m} \varphi(u).$$

We shall prove that

$$(5) \quad \sup_{u \in A_m} d(u, A) \rightarrow 0, \quad m \rightarrow \infty.$$

Let $\varepsilon > 0$ be fixed. If $\{u \in E : d(u, A) \geq \varepsilon\}$ is nonempty, the compactness of E implies that there exists $\delta_\varepsilon > 0$ such that

$$h(u) \leq \max_{u \in E} h(u) - \delta_\varepsilon, \quad d(u, A) \geq \varepsilon.$$

Choose $\tau \in A$. Since $\Delta_m \rightarrow 0$, there are points $z_m \in D_m$ with $z_m \rightarrow \tau$, and hence

$$\max_{z \in D_m} h(z) \geq h(z_m) \rightarrow \max_{u \in E} h(u), \quad m \rightarrow \infty.$$

For all sufficiently large m we therefore have

$$\max_{z \in D_m} h(z) > \max_{u \in E} h(u) - \delta_\varepsilon.$$

No maximizer over D_m can then have distance at least ε from A . If the displayed compact set is empty, the same conclusion is immediate. This proves the claim (5).

Since φ is continuous and equals c on the compact set A , uniform continuity combined with (5) yields

$$\frac{1}{|A_m|} \sum_{u \in A_m} \varphi(u) \rightarrow c, \quad m \rightarrow \infty,$$

which completes the proof. $\square$

Lemma 3.3. *Let (U_n, V_n) be jointly Gaussian random elements of $C(E) \times C(F)$, with marginal laws independent of n . If*

$$\sup_{u \in E, v \in F} |\text{Cov}(U_n(u), V_n(v))| \rightarrow 0, \quad n \rightarrow \infty,$$

then

$$(U_n, V_n) \Rightarrow (U, V) \quad \text{in } C(E) \times C(F),$$

where U and V are independent random elements with these marginal laws.

Proof. Tightness follows from the fixed marginal laws. For every finite collection of points, the corresponding Gaussian vectors have fixed marginal covariance blocks and off-diagonal blocks tending to zero. Hence their finite-dimensional distributions converge to product Gaussian distributions. Every subsequential weak limit therefore has the product finite-dimensional distributions, which determine the product Gaussian law on $C(E) \times C(F)$. $\square$

3.2. Covariance Price tools. We use the following Price-interpolation tools to study workload covariance asymptotics.

We first fix the notation and measurability conventions used below. For every compact metric space K fix a Borel map

$$\mathbf{a}_K : C(K) \longrightarrow K, \quad \mathbf{a}_K(h) \in \arg \max_{u \in K} h(u).$$

Such a selector exists by the measurable maximum theorem [3, Theorem 18.19]. Whenever the maximizer is unique, $\mathbf{a}_K(h)$ is that maximizer. If $(\theta, \omega) \mapsto Z^{(\theta)}(\omega)$ is a jointly measurable $C(K)$ -valued field, then $(\theta, \omega) \mapsto \mathbf{a}_K(Z^{(\theta)}(\omega))$ is jointly measurable.

On $K = [0, T]$, we take $\mathbf{a}_K$ to be the leftmost maximizer, and for a continuous path h on $[0, \infty)$ satisfying $h(t) \rightarrow -\infty$, we define $\mathbf{a}_\infty(h)$ to be its leftmost global maximizer. This map is measurable in the topology of uniform convergence on compact sets, since

$$\mathbf{a}_\infty(h) = \lim_{n \rightarrow \infty} \mathbf{a}_{[0, n]}(h|_{[0, n]}),$$

and the sequence is eventually constant once the compact interval contains the global maximizing set. All maximizers and atomic measures below are understood using these selectors.

Lemma 3.4. *If (X, Y) is a jointly Gaussian element of $C(E) \times C(F)$ with covariance $R(u, v) := \text{Cov}(X(u), Y(v))$, $u \in E, v \in F$ then almost surely R is constant on*

$$\arg \max_E X \times \arg \max_F Y.$$

Consequently, its value is independent of the measurable maximizer selections.

Proof. We first recall the Cameron–Martin tie observation. If Z is a continuous Gaussian process on a compact metric space K , with marginal RKHS $\mathcal{H}_Z$, then every fixed $h \in \mathcal{H}_Z$ is almost surely constant on $\arg \max_K Z$. Indeed, for $z \in C(K)$, let

$$\phi_z(r) := \max_{t \in K} \{z(t) + rh(t)\}.$$

This is a convex Lipschitz function of r , and the elementary one-dimensional Danskin formulas identify the right and left derivatives of ϕ_z as

$$\phi'_{z,+}(r) = \max_{t \in \arg \max(z+rh)} h(t), \quad \phi'_{z,-}(r) = \min_{t \in \arg \max(z+rh)} h(t).$$

Thus h is nonconstant on $\arg \max(z + rh)$ precisely when ϕ_z is not differentiable at r . For each z , this occurs at most at countably many r . The bad set in $C(K) \times \mathbb{R}$ is Borel, because the one-sided derivatives are limits of rational difference quotients. Fubini's theorem therefore ensures that there is a value r for which the bad event has probability zero for $Z + rh$. The Cameron–Martin theorem makes the laws of $Z + rh$ and Z equivalent, proving the observation.

Let $\mathcal{H}_X$ be the marginal RKHS of X . For each fixed $v \in F$, $R(\cdot, v)$ belongs to $\mathcal{H}_X$. To see this, project $Y(v) - \mathbb{E}\{Y(v)\}$ onto the closed Gaussian linear span of $\{X(u) - \mathbb{E}\{X(u)\} : u \in E\}$; under the canonical isometry from that span to $\mathcal{H}_X$, the projection corresponds exactly to $R(\cdot, v)$. The tie observation therefore shows that $R(\cdot, v)$ is constant on $\arg \max_E X$ almost surely. Intersecting these full-measure events over a countable dense subset of F and then using continuity of R yields the assertion simultaneously for every $v \in F$. The symmetric argument with the marginal RKHS of Y makes $R(u, \cdot)$ constant on $\arg \max_F Y$, simultaneously for every $u \in E$. Combining the two statements proves the lemma. $\square$

For a given jointly Gaussian element (X, Y) of $C(E) \times C(F)$ set $m_X(u) := \mathbb{E}\{X(u)\}$, $m_Y(v) := \mathbb{E}\{Y(v)\}$. Let X' and Y' be independent marginal copies of X and Y , independent of (X, Y) , and put

$$\begin{aligned} X^{(\theta)} &:= m_X + \sqrt{\theta}(X - m_X) + \sqrt{1 - \theta}(X' - m_X), \\ Y^{(\theta)} &:= m_Y + \sqrt{\theta}(Y - m_Y) + \sqrt{1 - \theta}(Y' - m_Y) \end{aligned}$$

for $\theta \in [0, 1]$. Also put

$$M_X := \max_{u \in E} X(u), \quad M_Y := \max_{v \in F} Y(v),$$

and let τ_X, τ_Y denote the corresponding selected maximizers.

Lemma 3.5. *If (X, Y) is a jointly Gaussian element of $C(E) \times C(F)$ then*

$$\text{Cov}(M_X, M_Y) = \int_0^1 \mathbb{E}\{R(\tau_{X^{(\theta)}}, \tau_{Y^{(\theta)}})\} d\theta.$$

Proof. Let us choose finite nets $D_m \subset E$ and $F_m \subset F$ with mesh tending to zero. For $\lambda > 0$, define

$$M_{m,\lambda}^X := \frac{1}{\lambda} \log \sum_{u \in D_m} e^{\lambda X(u)}, \quad M_{m,\lambda}^Y := \frac{1}{\lambda} \log \sum_{v \in F_m} e^{\lambda Y(v)}.$$

Apply lemma 3.1 to the centered finite-dimensional vectors $(X(u) - m_X(u))_{u \in D_m}$ and $(Y(v) - m_Y(v))_{v \in F_m}$. In these centered coordinates, the two functions are

$$z \mapsto \frac{1}{\lambda} \log \sum_{u \in D_m} e^{\lambda(m_X(u) + z_u)}, \quad w \mapsto \frac{1}{\lambda} \log \sum_{v \in F_m} e^{\lambda(m_Y(v) + w_v)}.$$

Their derivatives are the softmax weights:

$$\partial_{X(u)} M_{m,\lambda}^X = \frac{e^{\lambda X(u)}}{\sum_{z \in D_m} e^{\lambda X(z)}}, \quad u \in D_m,$$

and analogously on the Y -side. Applying lemma 3.1 therefore yields

$$\text{Cov}(M_{m,\lambda}^X, M_{m,\lambda}^Y) = \int_0^1 \mathbb{E} \left\{ \int_E \int_F R(u, v) \pi_{m,\lambda}^{X^{(\theta)}}(du) \pi_{m,\lambda}^{Y^{(\theta)}}(dv) \right\} d\theta.$$

For every $\theta \in (0, 1]$, $\text{Cov}(X^{(\theta)}(u), Y^{(\theta)}(v)) = \theta R(u, v)$. In view of lemma 3.4, R is constant on

$$(6) \quad \arg \max_E X^{(\theta)} \times \arg \max_F Y^{(\theta)}$$

almost surely. On the product net $D_m \times F_m$, the product measure $\pi_{m,\lambda}^{X^{(\theta)}} \otimes \pi_{m,\lambda}^{Y^{(\theta)}}$ is the softmax measure associated with the function $(u, v) \mapsto X^{(\theta)}(u) + Y^{(\theta)}(v)$, whose maximizing set is the displayed Cartesian product (6). Hence the generalized localization statement in lemma 3.2, applied with the test function R , yields the following convergence as $\lambda \rightarrow \infty$ and then $m \rightarrow \infty$

$$\int_E \int_F R(u, v) \pi_{m,\lambda}^{X^{(\theta)}}(du) \pi_{m,\lambda}^{Y^{(\theta)}}(dv) \longrightarrow R(\tau_{X^{(\theta)}}, \tau_{Y^{(\theta)}})$$

for $d\theta \otimes d\mathbb{P}$ -almost every (θ, ω) ; the single endpoint $\theta = 0$ is irrelevant. The absolute value is bounded by $\|R\|_\infty$, so dominated convergence applies on $[0, 1] \times \Omega$. On the left-hand side, the soft maxima first converge to the corresponding finite-net maxima, and the finite-net maxima converge to the compact suprema in L^2 (this follows from pathwise uniform continuity and the square-integrability of compact Gaussian suprema).

This completes the proof. $\square$

Lemma 3.6. *Let (X_r, Y_r) be jointly Gaussian in $C(E) \times C(F)$ with marginal laws independent of r . Assume that there exist continuous functions $g_1 : E \rightarrow \mathbb{R}$ and $g_2 : F \rightarrow \mathbb{R}$ and a function $b(r) > 0$ such that $b(r) \rightarrow 0$ as $r \rightarrow \infty$ and*

$$\sup_{u \in E, v \in F} \left| \frac{\text{Cov}(X_r(u), Y_r(v))}{b(r)} - g_1(u)g_2(v) \right| \rightarrow 0, \quad r \rightarrow \infty.$$

If the marginal maximizers are a.s. unique, then

$$\frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} \longrightarrow \mathbb{E}\{g_1(\tau_{\tilde{X}})\} \mathbb{E}\{g_2(\tau_{\tilde{Y}})\}, \quad r \rightarrow \infty,$$

where $\tilde{X}$ and $\tilde{Y}$ are independent random elements with the marginal laws of X_r and Y_r , respectively.

Proof. Let $R_r(u, v) := \text{Cov}(X_r(u), Y_r(v))$ and $K_r(u, v) := \frac{R_r(u, v)}{b(r)}$. In light of lemma 3.5 we have

$$\frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} = \int_0^1 \mathbb{E} \left\{ K_r(\tau_{X_r^{(\theta)}}(\theta), \tau_{Y_r^{(\theta)}}(\theta)) \right\} d\theta.$$

Uniform convergence allows us to replace K_r by $g_1 g_2$ at an error tending to zero as $r \rightarrow \infty$. Moreover we have

$$\text{Cov}(X_r^{(\theta)}(u), Y_r^{(\theta)}(v)) = \theta R_r(u, v).$$

Since $b(r) \rightarrow 0$, this cross-covariance block tends to zero uniformly on $E \times F$. Hence lemma 3.3 implies decoupling of the interpolated compact pair for each fixed θ . The map $x \mapsto (M_x, \tau_x)$ is continuous at every continuous path with a unique maximizer. Since g_1 and g_2 are continuous and bounded on their compact domains, we obtain

$$\mathbb{E} \left\{ g_1(\tau_{X_r^{(\theta)}}(\theta)) g_2(\tau_{Y_r^{(\theta)}}(\theta)) \right\} \longrightarrow \mathbb{E}\{g_1(\tau_{\tilde{X}})g_2(\tau_{\tilde{Y}})\} = \mathbb{E}\{g_1(\tau_{\tilde{X}})\} \mathbb{E}\{g_2(\tau_{\tilde{Y}})\}, \quad r \rightarrow \infty.$$

The limit is independent of θ , so dominated convergence on $[0, 1]$ establishes the claim. $\square$

We now fix the notation for the remainder of this section.

For each $r \geq 0$ let $X_r(t)$, $t \geq 0$ and $Y_r(t)$, $t \geq 0$ be jointly Gaussian real-valued processes with continuous sample paths a.s., whose marginal laws do not depend on r . Let X and Y be independent processes with these marginal laws, and put $m_X(t) := \mathbb{E}\{X_r(t)\}$, $m_Y(t) := \mathbb{E}\{Y_r(t)\}$ for their continuous means. Similarly to the already introduced X' and Y' , let X'_r and Y'_r be independent Gaussian processes, independent of each other and of (X_r, Y_r) , with laws equal to those of X_r and Y_r , respectively. Set

$$R_r(u, v) := \mathbb{Cov}(X_r(u), Y_r(v)), u \geq 0, v \geq 0$$

and for $\theta \in [0, 1]$ put

$$\begin{aligned} X_r^{(\theta)} &:= m_X + \sqrt{\theta}(X_r - m_X) + \sqrt{1-\theta}(X'_r - m_X), \\ Y_r^{(\theta)} &:= m_Y + \sqrt{\theta}(Y_r - m_Y) + \sqrt{1-\theta}(Y'_r - m_Y). \end{aligned}$$

For a measurable kernel K and finite signed atomic measures μ, ν on $[0, \infty)$, write

$$\mathcal{T}_K(\mu, \nu) := \int_0^\infty \int_0^\infty K(u, v) \mu(du) \nu(dv),$$

whenever the integral is well defined. Define the half-line and compact atoms

$$\begin{aligned} \Lambda_X^{r, \theta} &:= \delta_{\tau_{X_r^{(\theta)}}}, & \Lambda_{X, T}^{r, \theta} &:= \delta_{\tau_{X_r^{(\theta)}}(T)}, \\ \Lambda_Y^{r, \theta} &:= \delta_{\tau_{Y_r^{(\theta)}}}, & \Lambda_{Y, T}^{r, \theta} &:= \delta_{\tau_{Y_r^{(\theta)}}(T)}. \end{aligned}$$

Also let $b(r) > 0$, put $K_r(u, v) := \frac{R_r(u, v)}{b(r)}$, and define

$$\Delta_{X, T}^{r, \theta} := \Lambda_X^{r, \theta} - \Lambda_{X, T}^{r, \theta}, \quad \Delta_{Y, T}^{r, \theta} := \Lambda_Y^{r, \theta} - \Lambda_{Y, T}^{r, \theta}.$$

Lemma 3.7. *Suppose that the processes X and Y tend to $-\infty$ at infinity a.s. and have global maxima with finite second moments.*

Assume that there are nonnegative nondecreasing functions G_X, G_Y such that for every $r \geq 0$ we have

$$|R_r(u, v)| \leq G_X(u)G_Y(v), \quad u, v \geq 0,$$

and for each fixed r

$$\sup_{\theta \in [0, 1]} \mathbb{E}\left\{G_X(\tau_{X_r^{(\theta)}})G_Y(\tau_{Y_r^{(\theta)}})\right\} < \infty.$$

We obtain

$$(7) \quad \mathbb{Cov}(M_{X_r}, M_{Y_r}) = \int_0^1 \mathbb{E}\{\mathcal{T}_{R_r}(\Lambda_X^{r, \theta}, \Lambda_Y^{r, \theta})\} d\theta,$$

or equivalently

$$\mathbb{Cov}(M_{X_r}, M_{Y_r}) = \int_0^1 \mathbb{E}\left\{R_r(\tau_{X_r^{(\theta)}}, \tau_{Y_r^{(\theta)}})\right\} d\theta.$$

Assume in addition that

$$(8) \quad \lim_{T \rightarrow \infty} \limsup_{r \rightarrow \infty} \sup_{\theta \in [0, 1]} \mathbb{E}\left\{|\mathcal{T}_{K_r}(\Delta_{X, T}^{r, \theta}, \Lambda_Y^{r, \theta})| + |\mathcal{T}_{K_r}(\Lambda_X^{r, \theta}, \Delta_{Y, T}^{r, \theta})|\right\} = 0.$$

We also obtain

$$(9) \quad \lim_{T \rightarrow \infty} \limsup_{r \rightarrow \infty} \left| \frac{\mathbb{Cov}(M_{X_r}, M_{Y_r})}{b(r)} - \frac{\mathbb{Cov}(M_{X_r}(T), M_{Y_r}(T))}{b(r)} \right| = 0.$$

Proof. For a finite signed measure μ on $[0, \infty)$ and a nonnegative Borel weight w write

$$\|\mu\|_w := \int_0^\infty w(u) |\mu|(du).$$

We note that if x is a continuous path tending to $-\infty$ and its compact and global maximizing locations are chosen to be the leftmost ones, then for every nondecreasing w we have

$$(10) \quad \|\delta_{\tau_x(T)} - \delta_{\tau_x}\|_w \leq 2w(\tau_x)\mathbf{1}_{\{\tau_x > T\}}.$$

Indeed, on $\{\tau_x \leq T\}$ the leftmost compact and global maximizers agree. On $\{\tau_x > T\}$, one has $\tau_x(T) \leq T < \tau_x$, and monotonicity of w yields the bound.

Fix r and let $A < \infty$. Applying lemma 3.5 on $[0, A]^2$ yields

$$(11) \quad \text{Cov}(M_{X_r}(A), M_{Y_r}(A)) = \int_0^1 \mathbb{E}\{\mathcal{T}_{R_r}(\Lambda_{X,A}^{r,\theta}, \Lambda_{Y,A}^{r,\theta})\} d\theta.$$

Because the processes tend to $-\infty$, $M_{X_r}(A) \rightarrow M_{X_r}$ and $M_{Y_r}(A) \rightarrow M_{Y_r}$ a.s. as $A \rightarrow \infty$. Moreover we have

$$|M_{X_r}(A)| \leq |X_r(0)| + |M_{X_r}|, \quad |M_{Y_r}(A)| \leq |Y_r(0)| + |M_{Y_r}|.$$

The global maxima have finite second moments, so the left-hand side of (11) converges to $\text{Cov}(M_{X_r}, M_{Y_r})$. The kernel bound implies

$$|\mathcal{T}_{R_r}(\mu, \nu)| \leq \|\mu\|_{G_X} \|\nu\|_{G_Y}.$$

Using bilinearity, monotonicity of G_X, G_Y and (10) we obtain

$$\left| \mathcal{T}_{R_r}(\Lambda_{X,A}^{r,\theta}, \Lambda_{Y,A}^{r,\theta}) - \mathcal{T}_{R_r}(\Lambda_X^{r,\theta}, \Lambda_Y^{r,\theta}) \right| \leq C G_X(\tau_{X_r^{(\theta)}}) G_Y(\tau_{Y_r^{(\theta)}}) \left(\mathbf{1}_{\{\tau_{X_r^{(\theta)}} > A\}} + \mathbf{1}_{\{\tau_{Y_r^{(\theta)}} > A\}} \right).$$

Under the measurability convention above the displayed bound is jointly measurable in (θ, ω) . For every fixed θ , it tends to zero a.s. as $A \rightarrow \infty$, and hence it does so for $d\theta \otimes d\mathbb{P}$ -almost every (θ, ω) . Moreover it is bounded by

$$C G_X(\tau_{X_r^{(\theta)}}) G_Y(\tau_{Y_r^{(\theta)}}),$$

whose integral over $[0, 1] \times \Omega$ is at most

$$C \sup_{\theta \in [0, 1]} \mathbb{E} \left\{ G_X(\tau_{X_r^{(\theta)}}) G_Y(\tau_{Y_r^{(\theta)}}) \right\} < \infty.$$

Thus dominated convergence on $[0, 1] \times \Omega$ proves (7).

To establish (9), we apply (7) and the compact identity on $[0, T]^2$ and divide by $b(r)$. Since $K_r = R_r/b(r)$, we have

$$\frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} - \frac{\text{Cov}(M_{X_r}(T), M_{Y_r}(T))}{b(r)} = \int_0^1 \mathbb{E} \left\{ \mathcal{T}_{K_r}(\Lambda_X^{r,\theta}, \Lambda_Y^{r,\theta}) - \mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Lambda_{Y,T}^{r,\theta}) \right\} d\theta.$$

Using

$$\Lambda_X^{r,\theta} = \Lambda_{X,T}^{r,\theta} + \Delta_{X,T}^{r,\theta}, \quad \Lambda_Y^{r,\theta} = \Lambda_{Y,T}^{r,\theta} + \Delta_{Y,T}^{r,\theta},$$

bilinearity yields

$$\mathcal{T}_{K_r}(\Lambda_X^{r,\theta}, \Lambda_Y^{r,\theta}) - \mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Lambda_{Y,T}^{r,\theta}) = \mathcal{T}_{K_r}(\Delta_{X,T}^{r,\theta}, \Lambda_Y^{r,\theta}) + \mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Delta_{Y,T}^{r,\theta}).$$

Take absolute values and then $\limsup$ as r tends to ∞ . Finally let T tend to ∞ and apply (8) to obtain (9). $\square$

Theorem 3.8. *In the setting and notation of lemma 3.7, let $b(r) > 0$ with $b(r) \rightarrow 0$ as $r \rightarrow \infty$ and assume that (8) holds. Assume that for every $T > 0$ the maximizers of X and Y over $[0, T]$ are almost surely unique. If there exist continuous functions $g_1, g_2 : [0, \infty) \rightarrow \mathbb{R}$ such that for every $T > 0$*

$$(12) \quad \sup_{0 \leq u, v \leq T} |K_r(u, v) - g_1(u)g_2(v)| \rightarrow 0, \quad r \rightarrow \infty$$

and

$$(13) \quad g_1(\tau_X(T)) \rightarrow g_1(\tau_X), \quad g_2(\tau_Y(T)) \rightarrow g_2(\tau_Y), \quad T \rightarrow \infty,$$

in L^1 , then

$$(14) \quad \lim_{r \rightarrow \infty} \frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} = \mathbb{E}\{g_1(\tau_X)\} \mathbb{E}\{g_2(\tau_Y)\}.$$

Proof. Fix $T > 0$. In view of lemma 3.6 and (12) we have

$$\lim_{r \rightarrow \infty} \frac{\text{Cov}(M_{X_r}(T), M_{Y_r}(T))}{b(r)} = \mathbb{E}\{g_1(\tau_X(T))\} \mathbb{E}\{g_2(\tau_Y(T))\}.$$

Applying lemma 3.7 we obtain

$$\lim_{T \rightarrow \infty} \limsup_{r \rightarrow \infty} \left| \frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} - \frac{\text{Cov}(M_{X_r}(T), M_{Y_r}(T))}{b(r)} \right| = 0,$$

hence for every $T > 0$ we have

$$\begin{aligned} & \limsup_{r \rightarrow \infty} \left| \frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} - \mathbb{E}\{g_1(\tau_X)\} \mathbb{E}\{g_2(\tau_Y)\} \right| \\ & \leq \limsup_{r \rightarrow \infty} \left| \frac{\text{Cov}(M_{X_r}, M_{Y_r})}{b(r)} - \frac{\text{Cov}(M_{X_r}(T), M_{Y_r}(T))}{b(r)} \right| \\ & \quad + |\mathbb{E}\{g_1(\tau_X(T))\} \mathbb{E}\{g_2(\tau_Y(T))\} - \mathbb{E}\{g_1(\tau_X)\} \mathbb{E}\{g_2(\tau_Y)\}|. \end{aligned}$$

Letting $T \rightarrow \infty$ and using (13), we arrive at (14). $\square$

Remark 3.9. The preceding transfer theorem is stated in the product form needed for the queueing applications in this contribution. The same proof applies if the product kernel $g_1(u)g_2(v)$ is replaced by a continuous limiting kernel $\Gamma(u, v)$, provided the local uniform convergence, the normalized atom-cutoff condition, and the corresponding half-line integrability condition

$$\Gamma(\tau_X(T), \tau_Y(T)) \longrightarrow \Gamma(\tau_X, \tau_Y) \quad \text{in } L^1$$

are imposed for independent copies X and Y . We do not formulate this additional generality separately, since the models treated here have explicit product kernels.

3.3. Uniqueness and tail estimates.

Lemma 3.10. *Let T be a σ -compact metric space, and let $E \subset T$ be such that $T_0 := T \setminus E$ is σ -compact. Let $Z(t)$, $t \in T$, be a Gaussian process with continuous sample paths. If*

$$\text{Var}(Z(s) - Z(t)) > 0, \quad s, t \in T_0, \quad s \neq t,$$

then Z cannot attain its supremum over T_0 at two distinct points. If

$$\sup_{t \in E} Z(t) < \sup_{t \in T} Z(t) \quad \text{a.s.,}$$

and the supremum over T is attained a.s., then the supremum over T has an a.s. unique maximizer.

Proof. We apply the uniqueness criterion [8] on an increasing compact exhaustion of T_0 . The exceptional-set assumption rules out maximizers in E , and attainment over T ensures uniqueness of the global maximizer. $\square$

Lemma 3.11. *Let $c > 0$, $W(t) := B_H(t) - ct$, and*

$$M_W := \sup_{u \geq 0} W(u), \quad \tau_W := \arg \max_{u \geq 0} W(u).$$

We have $M_W < \infty$ a.s. The supremum of W over every compact interval $[0, T]$, and over $[0, \infty)$, is attained and has an a.s. unique maximizer. Moreover there exist constants $C, a > 0$ such that

$$\mathbb{P}\{\tau_W > L\} \leq Ce^{-aL^{2-2H}}, \quad L \geq 1.$$

Both M_W, τ_W belong to L^p for every $p < \infty$ and almost surely $W(t) \rightarrow -\infty$ as $t \rightarrow \infty$.

Proof. A dyadic Borell–TIS and Borell–Cantelli argument yields $B_H(u)/u \rightarrow 0$ as $u \rightarrow \infty$, a.s.; hence $W(u) \rightarrow -\infty$, a.s. Continuity ensures finiteness and attainment. On compact intervals, uniqueness follows from lemma 3.10, because

$$\text{Var}(B_H(u) - B_H(v)) = |u - v|^{2H} > 0, \quad u \neq v.$$

Global uniqueness follows by compact exhaustion.

For the tail, if $\tau_W > L$, then $\sup_{u \geq L} W(u) \geq 0$. On each dyadic block $[2^n L, 2^{n+1} L]$, by self-similarity and Borell–TIS

$$\mathbb{P} \left\{ \sup_{u \in [2^n L, 2^{n+1} L]} W(u) \geq 0 \right\} \leq \mathbb{P} \left\{ \sup_{u \in [1, 2]} B_H(u) \geq c(2^n L)^{1-H} \right\} \leq Ce^{-a(2^n L)^{2-2H}}.$$

Summing over $n \geq 0$ yields the stated tail and hence all moments of τ_W .

The assertion that $M_W \in L^p$ for every $p > 0$ follows directly from the tail asymptotics derived in [6]. $\square$

Lemma 3.12. *If (4) holds with the additional endpoint assumption $\lim_{t \rightarrow \infty} L(t) \log t = 0$ when $\alpha = 0$, then the following assertions hold with $b(t) := t^{-\alpha} L(t)$.*

(i) *For every $A > 0$ we have*

$$\lim_{t \rightarrow \infty} \sup_{|x| \leq A} \left| \frac{r_Z(t+x)}{b(t)} - \kappa \right| = 0.$$

For every $\eta \in (0, 1)$, there is $C_\eta < \infty$ such that

$$|r_Z(t+x)| \leq C_\eta b(t), \quad |x| \leq \eta t,$$

for all sufficiently large t .

(ii) *If $\alpha > 0$, then there exist $\beta < 2$ and $C < \infty$ such that*

$$V_X(t) \leq Ct^\beta, \quad t \geq 0.$$

If $\alpha = 0$, then for all large t we have

$$V_X(t) \leq Ct^2 L(t).$$

In both cases we have

$$\mathbb{E}\left\{ \sup_{0 \leq u \leq T} X(u) \right\} = o(T), \quad T \rightarrow \infty.$$

(iii) *For every $c > 0$, set $W(t) := X(t) - ct$. The maximum M_W is finite and attained a.s., and its maximizer τ_W is a.s. unique. The compact maximizers $\tau_W(T)$ are also a.s. unique, and $M_W, \tau_W \in L^p$ for every $p < \infty$. Moreover τ_W has a stretched-exponential tail when $\alpha > 0$, and a super-polynomial tail when $\alpha = 0$. Finally, $W(t) \rightarrow -\infty$ as $t \rightarrow \infty$, a.s.*

(iv) *For every $c > 0$, with $W(t) := X(t) - ct$, if*

$$G(x) := \left(\sup_{0 \leq y \leq x} V_X(y) \right)^{1/2},$$

then $G(\tau_W) \in L^p$ for every $p < \infty$, and for every $\eta \in (0, 1)$ we have

$$\lim_{r \rightarrow \infty} \frac{(\mathbb{E}\{G(\tau_W)^2 \mathbf{1}_{\{\tau_W > \eta r\}}\})^{1/2}}{b(r)} = 0.$$

Proof. Part (i) follows from the uniform convergence theorem and Potter bounds for regular variation; see, for example, [4, Sections 1.2 and 1.5]. Indeed, for fixed A , $t+x \sim t$ uniformly in $|x| \leq A$. For $|x| \leq \eta t$, one has

$$(1-\eta)t \leq t+x \leq (1+\eta)t$$

for all large t , and Potter bounds yield the stated uniform estimate.

We prove (ii). Recall that

$$V_X(t) = \mathbb{V}\text{ar}(X(t) - X(0)) = 2 \int_0^t (t-s) r_Z(s) ds, \quad t \geq 0.$$

First suppose that $\alpha > 0$. Potter bounds imply that for a suitable $\varepsilon > 0$ we have

$$|r_Z(s)| \leq C_\varepsilon s^{-\alpha+\varepsilon}, \quad s \geq 1.$$

If $\alpha \leq 1$, choose $\varepsilon > 0$ so small that

$$\beta := 2 - \alpha + \varepsilon < 2.$$

It follows that

$$\int_1^t (t-s) s^{-\alpha+\varepsilon} ds \leq Ct^\beta.$$

If $\alpha > 1$, choose $\varepsilon \in (0, \alpha - 1)$. The function $s^{-\alpha+\varepsilon}$ is integrable at infinity, and the same formula yields

$$V_X(t) \leq Ct.$$

Thus in this case we may take $\beta = 1$. Since r_Z is continuous on compact intervals, $V_X(t) \leq Ct^2$ for $0 \leq t \leq 1$. Enlarging C if necessary, we obtain

$$V_X(t) \leq Ct^\beta, \quad t \geq 0,$$

for some $\beta < 2$.

Now suppose that $\alpha = 0$. Here $b(t) = L(t)$, and the endpoint assumption implies $L(t) \rightarrow 0$ as $t \rightarrow \infty$. Since $r_Z(t) \sim \kappa L(t)$, Potter bounds imply that for every $\varepsilon \in (0, 1)$, there are s_0 and C_ε such that for all large t

$$|r_Z(s)| \leq C_\varepsilon L(t) \left(\frac{t}{s}\right)^\varepsilon, \quad s_0 \leq s \leq t.$$

The range $0 \leq s \leq s_0$ contributes at most Ct to $V_X(t)$. Therefore

$$\begin{aligned} V_X(t) &\leq Ct + C_\varepsilon L(t) t^\varepsilon \int_{s_0}^t (t-s) s^{-\varepsilon} ds \\ &\leq Ct + Ct^2 L(t). \end{aligned}$$

Since L is slowly varying and eventually positive, $tL(t) \rightarrow \infty$ as $t \rightarrow \infty$. Hence the term Ct is absorbed into $Ct^2 L(t)$, and so

$$V_X(t) \leq Ct^2 L(t)$$

for all large t .

It remains to prove the expected-supremum estimate. Define

$$\psi(T) := \begin{cases} T^{\beta-2}, & \alpha > 0, \\ L(T), & \alpha = 0. \end{cases}$$

This choice ensures

$$\psi(T) \log T \rightarrow 0, \quad T \rightarrow \infty.$$

The variance estimates just proved imply, after enlarging C , that

$$\sup_{0 \leq h \leq T} V_X(h) \leq CT^2 \psi(T), \quad T \geq 1.$$

For $\alpha = 0$, this uses Potter bounds once more: for every $\varepsilon \in (0, 2)$ we have

$$h^2 L(h) \leq CT^2 L(T) \left(\frac{h}{T}\right)^{2-\varepsilon} \leq CT^2 L(T), \quad h_0 \leq h \leq T,$$

and the bounded interval $0 \leq h \leq h_0$ is absorbed because $T^2 L(T) \rightarrow \infty$ as $T \rightarrow \infty$.

Let

$$d_X(s, t)^2 := \mathbb{E}\{(X(t) - X(s))^2\} = V_X(|t - s|)$$

be the canonical metric. Since

$$V_X(h) = \mathbb{V}\text{ar} \left(\int_0^h Z(u) du \right) \leq r_Z(0) h^2,$$

we have $d_X(s, t) \leq C|s - t|$. Moreover we have

$$\text{diam}_{d_X}([0, T]) \leq CT \sqrt{\psi(T)}.$$

Thus

$$N([0, T], d_X, \varepsilon) \leq 1 + \frac{CT}{\varepsilon}.$$

Dudley's entropy bound and the elementary estimate

$$\int_0^D \sqrt{\log \left(1 + \frac{A}{\varepsilon} \right)} d\varepsilon \leq CD \sqrt{\log \left(e + \frac{A}{D} \right)}$$

yield

$$\begin{aligned} \mathbb{E}\left\{\sup_{0 \leq u \leq T} X(u)\right\} &\leq C \int_0^{CT\sqrt{\psi(T)}} \sqrt{\log\left(1 + \frac{CT}{\varepsilon}\right)} d\varepsilon \\ &\leq CT\sqrt{\psi(T)} \sqrt{\log\left(e + \frac{1}{\sqrt{\psi(T)}}\right)} = o(T). \end{aligned}$$

This proves (ii).

We prove (iii). Since $b(t) \rightarrow 0$ as $t \rightarrow \infty$, we have $r_Z(t) \rightarrow 0$ as $t \rightarrow \infty$. Maruyama's ergodicity criterion for stationary Gaussian processes [9] implies that Z is ergodic. Since $Z(0) \in L^1$ and $\mathbb{E}\{Z(0)\} = 0$, the continuous-time ergodic theorem yields

$$\frac{X(t)}{t} = \frac{1}{t} \int_0^t Z(s) ds \rightarrow 0, \quad t \rightarrow \infty, \quad \text{a.s.}$$

Hence for every $c > 0$ we have

$$X(t) - ct \rightarrow -\infty, \quad t \rightarrow \infty, \quad \text{a.s.}$$

Since Z has continuous paths, X and $W(t) = X(t) - ct$ have continuous paths. Therefore $M_W < \infty$ a.s. and the supremum is attained.

We next prove uniqueness, so that τ_W is well defined before its tail is considered. Let μ be the spectral measure of Z . Since $r_Z(t) \rightarrow 0$ as $t \rightarrow \infty$, Maruyama's theorem, equivalently Wiener's criterion for stationary Gaussian processes, implies that μ has no atoms.

Suppose that $u \neq v$ and

$$\text{Var}(X(u) - X(v)) = 0.$$

The spectral representation therefore implies

$$\int_{\mathbb{R}} \left| \int_v^u e^{i\lambda x} dx \right|^2 \mu(d\lambda) = 0.$$

Hence μ is supported on the zero set of the Fourier transform of $\mathbf{1}_{[v,u]}$. This zero set is countable. Since μ is non-atomic and has positive total mass, this is impossible. Therefore

$$\text{Var}(X(u) - X(v)) > 0, \quad u \neq v.$$

In light of lemma 3.10, the maximizer over every compact interval is a.s. unique. Since the global maximum is attained, global uniqueness follows by compact exhaustion.

We next prove the tail bound for τ_W . For $T \geq 1$, put

$$S_T := \sup_{T \leq u \leq 2T} X(u).$$

Stationary increments imply

$$\begin{aligned} \mathbb{E}\{S_T\} &= \mathbb{E}\left\{X(T) + \sup_{0 \leq h \leq T} (X(T+h) - X(T))\right\} \\ &= \mathbb{E}\left\{\sup_{0 \leq h \leq T} X(h)\right\} = o(T). \end{aligned}$$

We also have

$$\sup_{T \leq u \leq 2T} \text{Var} X(u) \leq CT^2\psi(T)$$

for all large T . Borell–TIS therefore provides constants $a_0 > 0$ and $T_0 < \infty$ such that

$$\mathbb{P}\left\{\sup_{T \leq u \leq 2T} (X(u) - cu) \geq 0\right\} \leq \exp\left(-\frac{a_0}{\psi(T)}\right), \quad T \geq T_0.$$

If $\tau_W > x$, then $W(0) = 0$ implies

$$\sup_{u \geq x} W(u) \geq 0.$$

Consequently for $x \geq T_0$ we have

$$\begin{aligned} \mathbb{P}\{\tau_W > x\} &\leq \sum_{n=0}^{\infty} \mathbb{P}\left\{\sup_{2^n x \leq u \leq 2^{n+1}x} (X(u) - cu) \geq 0\right\} \\ &\leq \sum_{n=0}^{\infty} \exp\left(-\frac{a_0}{\psi(2^n x)}\right). \end{aligned}$$

For $1 \leq x < T_0$, the same bounds below hold after increasing the constants. Since $\psi(T) \log T \rightarrow 0$ as $T \rightarrow \infty$, for every $q > 0$ we have

$$\exp\left(-\frac{a_0}{\psi(T)}\right) \leq T^{-q-2}$$

for all large T . Hence

$$\mathbb{P}\{\tau_W > x\} \leq C_q x^{-q}, \quad x \geq 1,$$

for every $q > 0$. Thus $\tau_W \in L^p$ for every $p < \infty$. If $\alpha > 0$, then $\psi(T) = T^{\beta-2}$, so the same dyadic summation yields

$$\mathbb{P}\{\tau_W > x\} \leq C e^{-a x^{2-\beta}}, \quad x \geq 1,$$

which is the claimed stretched-exponential tail.

We now prove that $M_W \in L^p$ for every $p < \infty$. For $y \geq 1$ we have

$$\begin{aligned} \mathbb{P}\{M_W > y\} &\leq \mathbb{P}\left\{\sup_{0 \leq u \leq 1} (X(u) - cu) > y\right\} \\ &\quad + \sum_{n=0}^{\infty} \mathbb{P}\left\{\sup_{2^n \leq u \leq 2^{n+1}} (X(u) - cu) > y\right\}. \end{aligned}$$

The first term has a Gaussian tail by Borell–TIS on the compact interval $[0, 1]$. Put $T_n := 2^n$. For all large T_n , the same Borell–TIS argument yields

$$\mathbb{P}\left\{\sup_{T_n \leq u \leq 2T_n} (X(u) - cu) > y\right\} \leq \exp\left(-\frac{a_1(y + T_n)^2}{T_n^2 \psi(T_n)}\right).$$

The finitely many small blocks are absorbed into a compact Gaussian-tail term.

Split the remaining dyadic sum into $T_n \leq y$ and $T_n > y$. If $T_n \leq y$, then

$$T_n^2 \psi(T_n) \leq C y^2 \psi(y)$$

using Potter bounds in the endpoint case and therefore

$$\exp\left(-\frac{a_1(y + T_n)^2}{T_n^2 \psi(T_n)}\right) \leq \exp\left(-\frac{a_2}{\psi(y)}\right).$$

There are at most $C \log y$ such blocks, and since $\psi(y) \log y \rightarrow 0$ as $y \rightarrow \infty$, their total contribution is $O(y^{-q})$ for every $q > 0$. If $T_n > y$, then

$$\frac{(y + T_n)^2}{T_n^2 \psi(T_n)} \geq \frac{a_3}{\psi(T_n)}.$$

Using $\psi(T) \log T \rightarrow 0$ as $T \rightarrow \infty$ once more, we obtain

$$\exp\left(-\frac{a_3}{\psi(T_n)}\right) \leq T_n^{-q-2}$$

for all large n and hence

$$\sum_{T_n > y} \exp\left(-\frac{a_3}{\psi(T_n)}\right) \leq C_q y^{-q}.$$

Combining the compact block, the blocks $T_n \leq y$, and the blocks $T_n > y$, we obtain

$$\mathbb{P}\{M_W > y\} \leq C_q y^{-q}, \quad y \geq 1,$$

for every $q > 0$. Since $M_W \geq W(0) = 0$, this establishes $M_W \in L^p$ for every $p < \infty$.

This completes the proof of part (iii).

Finally we prove (iv). Since

$$|r_Z(t)| \leq r_Z(0), \quad t \in \mathbb{R},$$

we have

$$V_X(y) = \mathbb{V}\text{ar} \left(\int_0^y Z(s) ds \right) \leq r_Z(0)y^2$$

and therefore $G(x) \leq Cx$. The super-polynomial tail of τ_W proved above implies, by integration by parts, that

$$G(\tau_W) \in L^p, \quad p < \infty,$$

and for every $N > 0$ we have

$$\mathbb{E}\{G(\tau_W)^2 \mathbf{1}_{\{\tau_W > \eta r\}}\} \leq C_N r^{-N}.$$

Since L is slowly varying and eventually positive, we have

$$\frac{1}{L(r)} = o(r^\delta) \quad \text{for every } \delta > 0$$

implying thus

$$\frac{1}{b(r)} = \frac{r^\alpha}{L(r)} = o(r^{\alpha+\delta}) \quad \text{for every } \delta > 0.$$

Choosing N sufficiently large yields

$$\frac{(\mathbb{E}\{G(\tau_W)^2 \mathbf{1}_{\{\tau_W > \eta r\}}\})^{1/2}}{b(r)} \rightarrow 0, \quad r \rightarrow \infty.$$

This proves (iv) establishing the proo. □

4. PROOFS

For the non-Markovian proofs we use the following notation. For the relevant input process X set

$$(15) \quad \mathcal{X}(u) := X(0) - X(-u) - c_1 u, \quad \mathcal{Y}_r(v) := X(r) - X(r-v) - c_2 v, \quad u, v \geq 0.$$

By definition we have

$$Q_{W_1}(0) = M_{\mathcal{X}}, \quad Q_{W_2}(r) = M_{\mathcal{Y}_r},$$

and stationary increments imply that the marginal laws of $\mathcal{X}$ and $\mathcal{Y}_r$ are those of W_1 and W_2 , respectively. In the applications of Theorem 3.8 below compact uniqueness is used only for the compact product-kernel transfer step.

4.1. Proof of Theorem 2.1. First we note that the pair (Q_{W_1}, Q_{W_2}) is jointly stationary. Let $\mathcal{F}_0 := \sigma\{B(s) : s \leq 0\}$. Recall that $U_i = Q_{W_i}(0)$, $i = 1, 2$. The random variables U_1 and U_2 are $\mathcal{F}_0$ -measurable while the future increments $B(u) - B(0)$, $u \geq 0$ are independent of $\mathcal{F}_0$.

Proof of (2). Recalling $m_1 = \mathbb{E}\{U_1\} = 1/(2c_1)$, we note that

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \mathbb{E}\{(U_1 - m_1)Q_{W_2}(T)\} = \mathbb{E}\{(U_1 - m_1)\mathbb{E}\{Q_{W_2}(T) \mid \mathcal{F}_0\}\}.$$

We first identify the conditional dynamics of $Q_{W_2}(T)$ for $T \geq 0$. Note that

$$\begin{aligned} Q_{W_2}(T) &= \sup_{u \geq 0} (B(T) - B(T-u) - c_2 u) \\ &= \max \left\{ \sup_{0 \leq u \leq T} (B(T) - B(T-u) - c_2 u), \sup_{u \geq T} (B(T) - B(T-u) - c_2 u) \right\}. \end{aligned}$$

Substituting $s = T - u$ in the first supremum and $v = u - T$ in the second yields

$$Q_{W_2}(T) = \max \left\{ \sup_{0 \leq s \leq T} (B(T) - B(s) - c_2(T-s)), B(T) - B(0) - c_2 T + U_2 \right\}.$$

For $x \geq 0$ define $Q_t^{(2),x}$, $t \geq 0$ by

$$Q_t^{(2),x} = \max \left\{ \sup_{0 \leq s \leq t} (B(t) - B(s) - c_2(t-s)), B(t) - B(0) - c_2t + x \right\}.$$

For

$$Y_t^{(2),x} := x + B(t) - c_2t$$

a direct computation yields the pathwise identity

$$Q_t^{(2),x} = Y_t^{(2),x} + R_t^{(2),x}, \quad R_t^{(2),x} = \left(- \inf_{0 \leq u \leq t} Y_u^{(2),x} \right)^+.$$

Thus $Q_t^{(2),x}$, $t \geq 0$ is a reflected Brownian motion with drift $-c_2$ started from x and driven by the future Brownian increments. Conditionally on $\mathcal{F}_0$ the process $Q_{W_2}(T)$, $T \geq 0$ has the law of reflected Brownian motion with drift $-c_2$ started from U_2 and driven by the future Brownian increments.

For $a \geq 0$ define

$$H_T^{(2)}(a) := \mathbb{P}\{\tau_a^{(c_2)} \leq T\}.$$

Clearly we have

$$\mathbb{E}\{R_T^{(2),x}\} = \int_0^\infty \mathbb{P}\{R_T^{(2),x} > y\} dy = \int_0^\infty H_T^{(2)}(x+y) dy,$$

and therefore

$$\mathbb{E}\{Q_T^{(2),x}\} = x - c_2T + \int_0^\infty H_T^{(2)}(x+y) dy = x - c_2T + \int_x^\infty H_T^{(2)}(a) da$$

implying

$$\mathbb{E}\{Q_{W_2}(T) \mid \mathcal{F}_0\} = U_2 - c_2T + \int_{U_2}^\infty H_T^{(2)}(a) da.$$

Combining (4.1) with $\mathbb{E}\{U_1 - m_1\} = 0$ yields

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \mathbb{E}\{(U_1 - m_1)U_2\} + \mathbb{E}\left\{(U_1 - m_1) \int_{U_2}^\infty H_T^{(2)}(a) da\right\}.$$

We now justify the Fubini steps. Borell-TIS implies

$$\int_0^\infty H_T^{(2)}(a) da = \mathbb{E}\left\{\sup_{0 \leq u \leq T} (c_2u - B(u))\right\} < \infty.$$

Also $\mathbb{E}\{|U_1 - m_1|\} < \infty$. Fubini's theorem therefore yields

$$\mathbb{E}\left\{(U_1 - m_1) \int_{U_2}^\infty H_T^{(2)}(a) da\right\} = \int_0^\infty H_T^{(2)}(a) \mathbb{E}\{(U_1 - m_1) \mathbf{1}_{\{U_2 \leq a\}}\} da.$$

For $a \geq 0$ define

$$L_{12}(a) := \mathbb{E}\{(U_1 - m_1) \mathbf{1}_{\{U_2 > a\}}\}.$$

Since $\mathbb{E}\{U_1 - m_1\} = 0$ we have

$$\mathbb{E}\{(U_1 - m_1) \mathbf{1}_{\{U_2 \leq a\}}\} = -L_{12}(a).$$

Moreover we have

$$U_2 = \int_0^\infty \mathbf{1}_{\{U_2 > a\}} da,$$

and

$$\int_0^\infty |L_{12}(a)| da \leq \mathbb{E}\{|U_1 - m_1|U_2\} < \infty.$$

Another application of Fubini yields

$$\mathbb{E}\{(U_1 - m_1)U_2\} = \int_0^\infty L_{12}(a) da.$$

Combining these identities yields

$$\begin{aligned}\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) &= \int_0^\infty L_{12}(a) da - \int_0^\infty H_T^{(2)}(a) L_{12}(a) da \\ &= \int_0^\infty L_{12}(a) \{1 - H_T^{(2)}(a)\} da \\ &= \int_0^\infty \mathbb{E}\{(U_1 - m_1) \mathbf{1}_{\{U_2 > a\}}\} \mathbb{P}\{\tau_a^{(c_2)} > T\} da,\end{aligned}$$

which is the covariance identity (2).

Proof of (3). For $a > 0$ the hitting time $\tau_a^{(c_2)}$ has density (see [17])

$$f_a^{(2)}(u) = \frac{a}{\sqrt{2\pi}u^{3/2}} \exp\left(-\frac{(a - c_2 u)^2}{2u}\right), \quad u > 0.$$

Since

$$\int_0^\infty |L_{12}(a)| \mathbb{P}\{\tau_a^{(c_2)} > T\} da \leq \int_0^\infty |L_{12}(a)| da < \infty,$$

we may substitute the density and interchange the order of integration in (2) to obtain

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \int_0^\infty L_{12}(a) \int_T^\infty \frac{a}{\sqrt{2\pi}u^{3/2}} \exp\left(-\frac{(a - c_2 u)^2}{2u}\right) du da.$$

Using

$$-\frac{(a - c_2 u)^2}{2u} = -\frac{c_2^2 u}{2} + c_2 a - \frac{a^2}{2u},$$

we obtain

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \frac{1}{\sqrt{2\pi}} \int_T^\infty u^{-3/2} e^{-c_2^2 u/2} J_{12}(u) du,$$

where

$$J_{12}(u) := \int_0^\infty a e^{c_2 a} e^{-a^2/(2u)} L_{12}(a) da.$$

We next prove that $J_{12}(u)$ has a finite limit as $u \rightarrow \infty$. Since

$$|L_{12}(a)| \leq \mathbb{E}\{|U_1 - m_1| \mathbf{1}_{\{U_2 > a\}}\},$$

we have

$$\int_0^\infty a e^{c_2 a} |L_{12}(a)| da \leq \mathbb{E}\left\{|U_1 - m_1| \int_0^{U_2} a e^{c_2 a} da\right\} \leq C \mathbb{E}\{|U_1 - m_1|(1 + U_2)e^{c_2 U_2}\} < \infty,$$

for a finite constant C depending only on c_2 . The finiteness of the above follows by the combination of Hölder's inequality

$$\mathbb{E}\{|U_1 - m_1|(1 + U_2)e^{c_2 U_2}\} \leq (\mathbb{E}\{|U_1 - m_1|^p\})^{1/p} (\mathbb{E}\{(1 + U_2)^q e^{qc_2 U_2}\})^{1/q},$$

for any $q \in (1, 2)$ and $p = q/(q - 1)$, with the fact that $U_1 \sim \text{Exp}(2c_1)$, $U_2 \sim \text{Exp}(2c_2)$ (note that $qc_2 < 2c_2$). Therefore

$$a e^{c_2 a} L_{12}(a) \in L^1(0, \infty).$$

Dominated convergence yields

$$J_{12}(u) \rightarrow J_{12}(\infty) := \int_0^\infty a e^{c_2 a} L_{12}(a) da, \quad u \rightarrow \infty.$$

Let

$$A_2(T) := \int_T^\infty u^{-3/2} e^{-c_2^2 u/2} du.$$

We can write

$$\text{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \frac{J_{12}(\infty)}{\sqrt{2\pi}} A_2(T) + \frac{1}{\sqrt{2\pi}} \int_T^\infty u^{-3/2} e^{-c_2^2 u/2} \{J_{12}(u) - J_{12}(\infty)\} du.$$

The second term is $o(A_2(T))$ because

$$\sup_{u \geq T} |J_{12}(u) - J_{12}(\infty)| \rightarrow 0, \quad T \rightarrow \infty$$

and hence

$$\mathbb{Cov}(Q_{W_1}(0), Q_{W_2}(T)) = \frac{J_{12}(\infty) + o(1)}{\sqrt{2\pi}} A_2(T).$$

Finally we use

$$A_2(T) = \int_T^\infty u^{-3/2} e^{-c_2^2 u/2} du \sim \frac{2}{c_2^2} T^{-3/2} e^{-c_2^2 T/2}$$

to conclude that

$$\mathbb{Cov}(Q_{W_1}(0), Q_{W_2}(T)) \sim K_{12} T^{-3/2} e^{-c_2^2 T/2}$$

with

$$K_{12} := \frac{\sqrt{2/\pi}}{c_2^2} \int_0^\infty a e^{c_2 a} L_{12}(a) da.$$

It remains to calculate the constant K_{12} . The integrability estimate above permits another application of Fubini's theorem. Since

$$\int_0^x a e^{c_2 a} da = \frac{1}{c_2^2} \{e^{c_2 x} (c_2 x - 1) + 1\},$$

we obtain

$$\begin{aligned} J_{12}(\infty) &= \int_0^\infty a e^{c_2 a} \mathbb{E}\{(U_1 - m_1) \mathbf{1}_{\{U_2 > a\}}\} da \\ &= \mathbb{E}\left\{(U_1 - m_1) \int_0^{U_2} a e^{c_2 a} da\right\} \\ &= \frac{1}{c_2^2} \mathbb{E}\{(U_1 - m_1) \{e^{c_2 U_2} (c_2 U_2 - 1) + 1\}\} \\ &= \frac{1}{c_2^2} \mathbb{E}\{(U_1 - m_1) e^{c_2 U_2} (c_2 U_2 - 1)\}, \end{aligned}$$

where in the last equality we used that $\mathbb{E}\{U_1 - m_1\} = 0$.

Hence

$$K_{12} = \frac{\sqrt{2/\pi}}{c_2^4} \mathbb{E}\{(U_1 - m_1) e^{c_2 U_2} (c_2 U_2 - 1)\}.$$

We note that

$$\mathbb{E}\{(U_1 - m_1) e^{c_2 U_2} (c_2 U_2 - 1)\} = \mathbb{E}\{(U_1 - m_1) F_2(U_2)\} = \mathbb{E}\{U_1 F_2(U_2)\},$$

where

$$F_2(x) := e^{c_2 x} (c_2 x - 1), \quad x \geq 0$$

and the fact that $U_2 \sim \text{Exp}(2c_2)$ implies

$$\mathbb{E}\{F_2(U_2)\} = \int_0^\infty e^{c_2 x} (c_2 x - 1) 2c_2 e^{-2c_2 x} dx = 0.$$

For $c > 0$ write

$$M_c := \sup_{u \geq 0} (B(u) - cu).$$

Let $0 < \beta < \alpha$ and write

$$S := M_\alpha, \quad Z := M_\beta - M_\alpha.$$

Here $Z \geq 0$ and Jin, Pena, and Puha [7] identify the joint density of (S, Z) as

$$\begin{aligned} g_{\alpha, \beta}(x, z) &= 2\beta \sqrt{\alpha - \beta} \frac{x + 2z}{z^{3/2}} \varphi\left(\frac{(\alpha - \beta)x + \alpha z}{\sqrt{(\alpha - \beta)z}}\right) \\ &\quad + 4\beta(\alpha - 2\beta) e^{-2(\alpha - \beta)x - 2\beta z} \Phi\left(\frac{(\alpha - 2\beta)z - (\alpha - \beta)x}{\sqrt{(\alpha - \beta)z}}\right), \quad x, z > 0, \end{aligned}$$

where φ and Φ denote the standard normal density and distribution function.

If $c_1 < c_2$, take $\alpha = c_2$, $\beta = c_1$. We have

$$U_2 = M_{c_2} = S, \quad U_1 = M_{c_1} = S + Z.$$

Evaluating the integral we obtain

$$\mathbb{E}\{U_1 F_2(U_2)\} = \int_0^\infty \int_0^\infty (x+z) e^{c_2 x} (c_2 x - 1) g_{c_2, c_1}(x, z) dz dx = \frac{2(2c_2 - c_1)}{c_2^2}.$$

If $c_1 > c_2$, take $\alpha = c_1$, $\beta = c_2$. We have

$$U_1 = M_{c_1} = S, \quad U_2 = M_{c_2} = S + Z.$$

A similar calculation yields

$$\mathbb{E}\{U_1 F_2(U_2)\} = \int_0^\infty \int_0^\infty x e^{c_2(x+z)} (c_2(x+z) - 1) g_{c_1, c_2}(x, z) dz dx = \frac{2}{c_2} \left(1 - \frac{c_1 - c_2}{\sqrt{(c_1 - c_2)^2 + c_2^2}} \right).$$

If $c_1 = c_2 = c$, then $U_1 = U_2 =: U$, where $U \sim \text{Exp}(2c)$. Therefore we have

$$\mathbb{E}\{U_1 F_2(U_2)\} = 2c \int_0^\infty x e^{-cx} (cx - 1) dx = \frac{2}{c}.$$

We note that this agrees with $2(2c_2 - c_1)/c_2^2$ at $c_1 = c_2 = c$.

Consequently we obtain the strictly positive constant

$$K_{12} = \sqrt{\frac{2}{\pi}} \begin{cases} \frac{2(2c_2 - c_1)}{c_2^6}, & 0 < c_1 \leq c_2 \\ \frac{2}{c_2^5} \left(1 - \frac{c_1 - c_2}{\sqrt{(c_1 - c_2)^2 + c_2^2}} \right), & c_1 > c_2 \end{cases}.$$

This completes the proof. $\square$

4.2. Proof of Theorem 2.3. We use the notation (15) with $X = B_H$ and $H \in (0, 1) \setminus \{1/2\}$. Let $A_H := H(2H - 1)$ and $b(r) := r^{2H-2}$. For $0 \leq u, v \leq T$ and $r > T$ the definition yields

$$\begin{aligned} R_r(u, v) &:= \text{Cov}(\mathcal{X}(u), \mathcal{Y}_r(v)) = \frac{1}{2} \{ (r+u)^{2H} + |r-v|^{2H} - |r+u-v|^{2H} - r^{2H} \} \\ (16) \quad &= A_H \int_0^u \int_0^v (r + \alpha - \beta)^{2H-2} d\beta d\alpha. \end{aligned}$$

Consequently for every $T \in (0, \infty)$ the following convergence holds uniformly on $0 \leq u, v \leq T$

$$\frac{R_r(u, v)}{b(r)} \longrightarrow A_H uv, \quad r \rightarrow \infty.$$

We shall apply Theorem 3.8. Let $\mathcal{X}^{(\theta)}$ and $\mathcal{Y}_r^{(\theta)}$ be the Price-interpolated processes from Lemma 3.7. For every $\theta \in [0, 1]$ their marginal laws are those of W_1 and W_2 . Lemma 3.11 establishes convergence to $-\infty$, compact and global uniqueness, finite second moments of the maxima and all moments of the maximizers.

Cauchy's inequality yields

$$(17) \quad |R_r(u, v)| \leq u^H v^H, \quad u, v \geq 0.$$

Hence we may take $G_X(u) = u^H$ and $G_Y(v) = v^H$. If

$$\xi := \tau_{\mathcal{X}^{(\theta)}}, \quad \zeta := \tau_{\mathcal{Y}_r^{(\theta)}},$$

then ξ and ζ have marginal laws τ_{W_1} and τ_{W_2} and therefore

$$\sup_{r, \theta} \mathbb{E}\{G_X(\xi) G_Y(\zeta)\} \leq (\mathbb{E}\{\tau_{W_1}^{2H}\})^{1/2} (\mathbb{E}\{\tau_{W_2}^{2H}\})^{1/2} < \infty.$$

We now verify condition (8). We recall that $K_r := R_r/b(r)$ and observe that if the compact and global atoms differ on the X -side, then $\xi > T$ and the compact atom location is bounded by ξ ; the analogous statement holds on the Y -side with ζ .

Fix $\eta \in (0, 1)$. On $0 \leq u, v \leq \eta r$ the integral representation (16) yields

$$|K_r(u, v)| \leq C_\eta uv \leq C_\eta(1+u)(1+v).$$

On $\{\xi \leq \eta r, \zeta \leq \eta r\}$ we therefore obtain

$$|\mathcal{T}_{K_r}(\Delta_{X,T}^{r,\theta}, \Lambda_Y^{r,\theta})| + |\mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Delta_{Y,T}^{r,\theta})| \leq C_\eta(1+\xi)(1+\zeta) (\mathbf{1}_{\{\xi > T\}} + \mathbf{1}_{\{\zeta > T\}}).$$

Hölder's inequality and the moment bounds in Lemma 3.11 show that the expectation of the right-hand side tends to zero as $T \rightarrow \infty$ uniformly in r and θ .

On the complementary far event (17) yields

$$|K_r(u, v)| \leq r^{2-2H} u^H v^H.$$

Thus the far contribution is bounded by

$$C r^{2-2H} \mathbb{E}\{\xi^H \zeta^H (\mathbf{1}_{\{\xi > \eta r\}} + \mathbf{1}_{\{\zeta > \eta r\}})\}.$$

Cauchy's inequality, the fixed marginal laws and the maximizing-time tail estimate in Lemma 3.11 imply that this bound tends to zero as $r \rightarrow \infty$. Hence (8) holds.

Finally $\tau_{W_i}(T) \rightarrow \tau_{W_i}$ in L^1 . Indeed the compact maximizer is bounded by the global maximizer and equals it once $T \geq \tau_{W_i}$, while $\tau_{W_i} \in L^1$. Applying Theorem 3.8 with

$$g_1(u) = u, \quad g_2(v) = A_H v,$$

yields

$$\lim_{r \rightarrow \infty} \frac{\text{Cov}(Q_{W_1}(0), Q_{W_2}(r))}{r^{2H-2}} = A_H \mathbb{E}\{\tau_{W_1}\} \mathbb{E}\{\tau_{W_2}\}.$$

The moment factors are positive by Remark 2.6 and $A_H \neq 0$ because $H \neq 1/2$. This completes the proof. $\square$

4.3. Proof of Theorem 2.4. Set next

$$b(r) := r^{-\alpha} L(r).$$

Here $b(r) > 0$ for all large r and $b(r) \rightarrow 0$ as $r \rightarrow \infty$. In the notation (15) the cross-covariance kernel is

$$R_r(u, v) := \text{Cov}(\mathcal{X}(u), \mathcal{Y}_r(v)) = \int_{-u}^0 \int_{r-v}^r r_Z(t-s) dt ds = \int_0^u \int_0^v r_Z(r+s-t) dt ds.$$

Lemma 3.12 implies that uniformly on every compact square

$$\frac{R_r(u, v)}{b(r)} \longrightarrow \kappa uv, \quad r \rightarrow \infty.$$

We shall apply Theorem 3.8. As in the proof of Theorem 2.3 let $\mathcal{X}^{(\theta)}$ and $\mathcal{Y}_r^{(\theta)}$ be the Price-interpolated processes from Lemma 3.7. For every $\theta \in [0, 1]$ their marginal laws are those of W_1 and W_2 . Thus Lemma 3.12 establishes convergence to $-\infty$, compact and global uniqueness, finite second moments of the maxima and the required moment bounds for the maximizers.

Let

$$V_X(h) := \mathbb{V}\text{ar}(X(h) - X(0)), \quad G(x) := \left(\sup_{0 \leq y \leq x} V_X(y) \right)^{1/2}.$$

Cauchy's inequality implies

$$(18) \quad |R_r(u, v)| \leq \sqrt{V_X(u)} \sqrt{V_X(v)} \leq G(u)G(v).$$

The function G is nonnegative and nondecreasing. If

$$\xi := \tau_{\mathcal{X}^{(\theta)}}, \quad \zeta := \tau_{\mathcal{Y}_r^{(\theta)}},$$

then ξ and ζ have marginal laws τ_{W_1} and τ_{W_2} . Hence

$$\sup_{r, \theta} \mathbb{E}\{G(\xi)G(\zeta)\} \leq (\mathbb{E}\{G(\tau_{W_1})^2\})^{1/2} (\mathbb{E}\{G(\tau_{W_2})^2\})^{1/2} < \infty$$

by Lemma 3.12.

We now verify condition (8). If the compact and global atoms differ on the X -side, then $\xi > T$ and the compact atom location is bounded by ξ ; similarly on the Y -side.

Fix $\eta \in (0, 1)$. For all large r we obtain from Lemma 3.12

$$|r_Z(r+x)| \leq C_\eta b(r), \quad |x| \leq \eta r.$$

Thus for $0 \leq u, v \leq \eta r$ we have

$$|K_r(u, v)| \leq C_\eta uv \leq C_\eta(1+u)(1+v).$$

On $\{\xi \leq \eta r, \zeta \leq \eta r\}$ this yields

$$|\mathcal{T}_{K_r}(\Delta_{X,T}^{r,\theta}, \Lambda_Y^{r,\theta})| + |\mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Delta_{Y,T}^{r,\theta})| \leq C_\eta(1+\xi)(1+\zeta) (\mathbf{1}_{\{\xi > T\}} + \mathbf{1}_{\{\zeta > T\}}).$$

Hölder's inequality and Lemma 3.12 show that the expectation of the right-hand side tends to zero as $T \rightarrow \infty$ uniformly in r and θ .

On the complementary far event (18) yields

$$|K_r(u, v)| \leq \frac{G(u)G(v)}{b(r)}.$$

The far contribution is therefore bounded by

$$\frac{C}{b(r)} G(\xi)G(\zeta) (\mathbf{1}_{\{\xi > \eta r\}} + \mathbf{1}_{\{\zeta > \eta r\}}).$$

Cauchy's inequality, the fixed marginal laws and Lemma 3.12(iv) show that its expectation tends to zero as $r \rightarrow \infty$. Hence (8) holds.

Finally $\tau_{W_i}(T) \rightarrow \tau_{W_i}$ in L^1 since the compact maximizer is bounded by the global maximizer and eventually equals it, while $\tau_{W_i} \in L^1$ by Lemma 3.12. Applying Theorem 3.8 with

$$g_1(u) = u, \quad g_2(v) = \kappa v,$$

yields

$$\lim_{r \rightarrow \infty} \frac{\text{Cov}(Q_{W_1}(0), Q_{W_2}(r))}{r^{-\alpha} L(r)} = \kappa \mathbb{E}\{\tau_{W_1}\} \mathbb{E}\{\tau_{W_2}\}.$$

Since $\kappa \neq 0$ and the moment factors are positive by Remark 2.6, the limiting constant is nonzero, which proves the asserted asymptotic equivalence. This completes the proof. $\square$

4.4. Proof of Theorem 2.5. We shall apply Theorem 3.8. The variogram of $X(t) = \int_0^t Z(s) ds$ is

$$V_X(h) := \text{Var}(X(h) - X(0)) = \frac{2\sigma^2}{\lambda} h - \frac{2\sigma^2}{\lambda^2} (1 - e^{-\lambda h}), \quad h \geq 0.$$

The function V_X is nonnegative and strictly increasing on $(0, \infty)$ and satisfies

$$V_X(h) > 0, \quad h > 0, \quad V_X(h) \leq C(1+h).$$

Thus for $W(t) = X(t) - ct$ compact uniqueness of the maximizers follows from Lemma 3.10.

The result of [18] implies

$$X(u) - cu \rightarrow -\infty, \quad u \rightarrow \infty, \quad \text{a.s.}$$

Continuity now implies that the global maximum is finite and attained. Compact uniqueness and compact exhaustion imply that its maximizer τ_W is a.s. unique.

For $n \geq 1$ put

$$S_n := \sup_{n \leq u \leq n+1} X(u), \quad m_* := \mathbb{E}\left\{ \sup_{0 \leq u \leq 1} X(u) \right\} < \infty.$$

Stationary increments imply $\mathbb{E}\{S_n\} = m_*$ while monotonicity of V_X yields

$$\sup_{n \leq u \leq n+1} \text{Var} X(u) = V_X(n+1) = \frac{2\sigma^2}{\lambda} n + O(1).$$

Thus the Borell–TIS inequality [2] yields

$$\mathbb{P}\{S_n \geq x\} \leq \exp\left(-\frac{(x - m_*)^2_+}{2V_X(n+1)}\right), \quad x \in \mathbb{R}.$$

We next verify the required moment bound for the maximum. For all sufficiently large y we have

$$\mathbb{P}\{M_W > y\} \leq \mathbb{P}\left\{\sup_{0 \leq u \leq 1} (X(u) - cu) > y\right\} + \sum_{n=1}^{\infty} \mathbb{P}\{S_n > y + cn\}.$$

The first term has a Gaussian tail. Since $V_X(n+1) \leq Cn$ for $n \geq 1$, Borell–TIS yields constants $a_1, a_2, C > 0$ such that

$$\mathbb{P}\{S_n > y + cn\} \leq Ce^{-a_1 y - a_2 n}, \quad n \geq 1.$$

Indeed for $y \geq 2m_*$ we have

$$\frac{(y + cn - m_*)^2}{2V_X(n+1)} \geq a_1 y + a_2 n.$$

Summing over n yields

$$\mathbb{P}\{M_W > y\} \leq Ce^{-a_1 y}, \quad y \geq 1,$$

after changing the constants. In particular we have

$$M_W \in L^p, \quad p < \infty.$$

It remains to estimate the maximizing location. Since $W(0) = 0$ the event $\{\tau_W > T\}$ implies

$$\sup_{u \geq T} (X(u) - cu) \geq 0.$$

Thus for all sufficiently large T we obtain

$$\mathbb{P}\{\tau_W > T\} \leq \sum_{n=\lfloor T \rfloor}^{\infty} \mathbb{P}\{S_n \geq cn\} \leq C_\varepsilon \exp\left[-\left(\frac{c^2 \lambda}{4\sigma^2} - \varepsilon\right) T\right]$$

for every sufficiently small $\varepsilon > 0$. Writing

$$\iota(c) := \frac{c^2 \lambda}{4\sigma^2},$$

we obtain for every $q \in (0, \iota(c))$

$$\mathbb{E}\{e^{q\tau_W}\} = 1 + q \int_0^\infty e^{qt} \mathbb{P}\{\tau_W > t\} dt < \infty.$$

Applying these conclusions with $c = c_i$ for $i = 1, 2$ establishes all the required one-dimensional properties for the two marginal workloads. Moreover $c_2 > 2\sigma$ implies

$$\iota(c_2) > \lambda,$$

and hence

$$\mathbb{E}\{e^{\lambda\tau_{W_2}}\} < \infty.$$

We note that no exponential moment of τ_{W_1} at rate λ is needed below.

In the notation (15) we have

$$\mathcal{X}(u) = \int_{-u}^0 Z(a) da - c_1 u, \quad \mathcal{Y}_r(v) = \int_{r-v}^r Z(b) db - c_2 v.$$

The cross-covariance kernel satisfies

$$R_r(u, v) := \text{Cov}(\mathcal{X}(u), \mathcal{Y}_r(v)) = \int_0^u \int_0^v \sigma^2 e^{-\lambda|r+\alpha-\beta|} d\beta d\alpha,$$

and for $0 \leq u, v \leq T$ and $r > T$ we obtain

$$R_r(u, v) = \frac{\sigma^2}{\lambda^2} e^{-\lambda r} (1 - e^{-\lambda u})(e^{\lambda v} - 1).$$

Thus with

$$b(r) := e^{-\lambda r}, \quad g_1(u) := 1 - e^{-\lambda u}, \quad g_2(v) := \frac{\sigma^2}{\lambda^2} (e^{\lambda v} - 1),$$

we have $R_r/b(r) \rightarrow g_1 g_2$ uniformly on every compact square as $r \rightarrow \infty$.

It remains to verify condition (8). The interpolated marginal laws are those of W_1 and W_2 so the one-dimensional properties proved above apply to them.

Cauchy's inequality implies

$$|R_r(u, v)| \leq \sqrt{V_X(u)} \sqrt{V_X(v)}.$$

Since V_X is nonnegative and nondecreasing the function

$$G(x) := \sqrt{V_X(x)}$$

is nonnegative and nondecreasing. Moreover $V_X(x) \leq C(1+x)$ so the domination condition holds with this choice of G and the corresponding product-integrability follows from the finite moments of the maximizers.

Put $K_r := R_r/b(r)$. For $h > 0$, $u, v \geq 0$ and $\beta \geq 0$ we have

$$e^{\lambda h} \int_0^u e^{-\lambda|h+\alpha-\beta|} d\alpha \leq \frac{2}{\lambda} e^{\lambda\beta}.$$

Consequently we obtain

$$|K_r(u, v)| \leq C e^{\lambda v}, \quad u, v \geq 0.$$

Let

$$\xi := \tau_{\mathcal{X}^{(\theta)}}, \quad \zeta := \tau_{\mathcal{Y}^{(\theta)}}.$$

If the compact and global atoms differ on the X -side, then $\xi > T$; if they differ on the Y -side, then $\zeta > T$. Hence

$$|\mathcal{T}_{K_r}(\Delta_{X,T}^{r,\theta}, \Lambda_Y^{r,\theta})| + |\mathcal{T}_{K_r}(\Lambda_{X,T}^{r,\theta}, \Delta_{Y,T}^{r,\theta})| \leq C e^{\lambda\zeta} (\mathbf{1}_{\{\xi > T\}} + \mathbf{1}_{\{\zeta > T\}}).$$

The variables ζ have the common law τ_{W_2} and $\mathbb{E}\{e^{\lambda\tau_{W_2}}\} < \infty$. Therefore

$$\mathbb{E}\{e^{\lambda\zeta} \mathbf{1}_{\{\zeta > T\}}\} \rightarrow 0, \quad T \rightarrow \infty.$$

For the term with $\{\xi > T\}$ the family $\{e^{\lambda\zeta} : r \geq 0, \theta \in [0, 1]\}$ is uniformly integrable, while $\mathbb{P}\{\xi > T\} = \mathbb{P}\{\tau_{W_1} > T\} \rightarrow 0$ as $T \rightarrow \infty$. Thus

$$\sup_{r,\theta} \mathbb{E}\{e^{\lambda\zeta} \mathbf{1}_{\{\xi > T\}}\} \rightarrow 0, \quad T \rightarrow \infty.$$

This proves (8).

Finally the compact maximizers satisfy $0 \leq \tau_{W_i}(T) \leq \tau_{W_i}$ and eventually equal τ_{W_i} . Consequently we obtain

$$1 - e^{-\lambda\tau_{W_1}(T)} \rightarrow 1 - e^{-\lambda\tau_{W_1}}, \quad T \rightarrow \infty,$$

by dominated convergence and

$$e^{\lambda\tau_{W_2}(T)} - 1 \rightarrow e^{\lambda\tau_{W_2}} - 1, \quad T \rightarrow \infty,$$

in L^1 by $\mathbb{E}\{e^{\lambda\tau_{W_2}}\} < \infty$. Applying Theorem 3.8 yields

$$\lim_{r \rightarrow \infty} \frac{\text{Cov}(Q_{W_1}(0), Q_{W_2}(r))}{e^{-\lambda r}} = \frac{\sigma^2}{\lambda^2} \mathbb{E}\{1 - e^{-\lambda\tau_{W_1}}\} \mathbb{E}\{e^{\lambda\tau_{W_2}} - 1\}.$$

The two expectation factors are positive by Remark 2.6. This completes the proof. $\square$

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